Testing Compositional Faithfulness: Causal Interventions on Sparse Autoencoder Features in a Combinatorial Bijection Transformer

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https://github.com/RaggedR/rsk-transformer

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Abstract

Gauderis et al. recently proposed *compositional interpretability*, a category-theoretic framework that formalises when a mechanistic explanation is both faithful to a model's internal computation and compositionally consistent. We provide the first empirical test of this framework by performing causal feature interventions on sparse autoencoder (SAE) decompositions of a transformer trained to invert combinatorial bijections. We study two tasks: permutation RSK (where multiple algorithms exist) and cylindric plane partitions (where the Burge local rule is the only known algorithm). Zeroing individual SAE features reveals a striking contrast: in the cylindric model, 29% of causally important features are also compositionally faithful (causally necessary *and* correlated with specific growth diagram faces), with the three most impactful features all showing $> 3\times$ lift on identified faces. In the permutation model, only 2% of causal features are compositionally faithful—the per-step bumping path correlations reported in prior work are almost entirely epiphenomenal. This contrast supports the Weak Principle of Explanatory Optimism selectively: compositional faithfulness emerges when the task constrains the model to a unique algorithm, but not when multiple algorithmic strategies are available.

1 Introduction

Mechanistic interpretability aims to reverse-engineer neural networks into human-understandable components. The standard approach—decompose, describe, validate (**?**)—relies on correlational evidence: a feature "looks like" it encodes a concept, an attention head "seems to" perform a specific computation. But correlation is not causation. A sparse autoencoder (SAE) feature that fires on inputs matching a particular algorithmic step may be decorative rather than functional, tracking a pattern without contributing to the model's actual computation.

Gauderis et al. (**?**) recently proposed *compositional interpretability*, a category-theoretic framework that addresses this gap. They define a *compositional interpretation* as a pair of mappings—one structural (I_S, labelling syntax), one behavioural (I_C, labelling semantics)—that must *commute*: $I_S = I_C \circ \cdot$. When this commutativity holds, the structural label assigned to a component agrees with its behavioural role. They call this *compositional faithfulness*.

We provide the first empirical test of this framework. Our test bed is a 1.2M-parameter transformer trained to invert combinatorial bijections (**?**), where the ground-truth algorithms are known exactly. We study two tasks:

1. **Permutation RSK** ($n = 10$, 100% test accuracy): inverting the Robinson–Schensted–Knuth correspondence, where two fundamentally different algorithms exist—Schensted's sequential bumping and Fomin's parallel growth diagrams via local rules.

2. **Cylindric plane partitions** (profile $(1, 0, 1, 0, 1, 0, 1, 0)$, 99.98% accuracy): inverting a cylindric growth diagram bijection where the *only* known algorithm is the Burge local rule applied face-by-face (**?**).

For each task, prior work (**?**) trained SAEs on the transformer's residual stream and found features that *correlate* with specific algorithmic steps (bumping path steps for RSK, growth diagram faces for cylindric). We extend this by asking: are these features *causally necessary*? We zero each SAE feature individually, patch the reconstructed activation back into the model's forward pass, and measure accuracy drop.

The results reveal a sharp contrast between the two tasks, with direct implications for the compositional interpretability framework.

2 Background

2.1 Compositional Interpretability

We briefly summarise the framework of **?**. A *compositional model* $M = (D, \mathbf{S}, \mathbf{C}, \cdot)$ consists of a string diagram D in a syntactic monoidal category $\mathbf{S}$, a semantic monoidal category $\mathbf{C}$, and a representation functor $\cdot : \mathbf{S} \to \mathbf{C}$ mapping structure to behaviour. A *compositional interpretation* is a pair of maps $I_S : \mathbf{S} \to H$ (syntactic grounding) and $I_C : \mathbf{C} \to H$ (semantic grounding) into a space H of human-interpretable terms, satisfying the commutativity condition $I_S = I_C \circ \cdot$.

Definition 1 (Compositional faithfulness, (**?**, Def. 3.1))**.** *A compositional interpretation $I_S : \mathbf{S} \to H$ is* compositionally faithful *if there exists a reconstruction $I_C^* : H \to \mathbf{C}$ such that for every generator f of D, the semantic distortion $d(f, I_C^*(I_S(f))) \leq \epsilon$.*

Gauderis et al. classify existing mechanistic interpretability methods into a five-level *refinement hierarchy* based on how the refinement functor $R : \mathbf{S} \to \mathbf{S}'$ is constrained. SAE-based analysis operates at Level 2 (piecewise): each component is refined in isolation, with a per-component fidelity bound.

Conjecture 1 (Weak Principle of Explanatory Optimism, (**?**, Conj. 4.3))**.** *Most important behaviour of an intelligence with implicit explanatory knowledge lies within some human-understandable complexity class.*

If this conjecture holds, syntactic compression reliably surfaces interpretable structure, because the regularities the model exploits align with human-understandable concepts.

2.2 The RSK Transformer

The RSK transformer (**?**) is a 1.2M-parameter encoder-only transformer that learns to invert combinatorial bijections. Given a pair of tableaux (P, Q) encoded as structured 2D tokens (value + row + column + tableau identity embeddings, summed into 128-d vectors), the model predicts the original permutation σ. The same architecture handles permutation RSK, matrix RSK, reverse plane partitions (Hillman–Grassl), and cylindric plane partitions, achieving near-perfect accuracy on all tasks.

Prior interpretability analysis trained TopK SAEs ($d_{\text{dict}} = 1024$, $k = 32$) on residual stream activations at each of the 6 transformer layers. Two families of features were identified:

- **Permutation RSK**: *insertion-order detectors* with monotonically increasing per-step lift, and *step-specific Q-entry locators* that fire on particular (v, r, c) combinations in Q.
- **Cylindric**: *growth diagram face detectors* that fire on the partition triples $(i{-}1, i, i{+}1)$ involved in specific Burge local rule applications, at 3.0–3.3× lift.

These findings are correlational. Our experiment tests whether they are causal.

3 Experiment Design

3.1 Causal Feature Ablation

For a trained model M, a trained SAE at layer ℓ, and a feature index j, we define the *single-feature ablation* as follows:

1. Run the forward pass up to layer ℓ.
2. Intercept the residual stream activation $h \in \mathbb{R}^{b \times t \times d}$.
3. Encode through the SAE: normalise, apply encoder, apply TopK sparsity to obtain sparse activations $z \in \mathbb{R}^{d_{\text{dict}}}$.
4. Set $z_j = 0$ (the intervention).
5. Decode back to $\hat{h}$ and substitute into the forward pass.
6. Measure downstream accuracy.

The *causal importance* of feature j is $\Delta_j = a_{\text{SAE}} - a_j$, where a_{SAE} is the accuracy with full SAE reconstruction (no features zeroed) and a_j is the accuracy with feature j zeroed. Using the SAE baseline rather than the original model isolates the effect of the single feature from the SAE reconstruction error.

3.2 Cross-Referencing Causal and Correlational Evidence

For each feature, we have two independent measurements:

- **Correlational**: the *per-step lift* (permutation) or *per-face lift* (cylindric)—how much more likely the feature is to fire on tokens involved in a specific algorithmic step compared to its base rate.
- **Causal**: Δ_j, the accuracy drop from zeroing the feature.

A feature is *compositionally faithful* (in the sense of **?**) if it scores high on *both* measures: the structural label assigned by the correlational analysis (I_S) agrees with the feature's functional role as revealed by the causal intervention (I_C). We operationalise this as $\Delta_j > 0.1\%$ (causal) and lift $> 1.5\times$ (correlated).

3.3 Setup

- **Permutation RSK**: model `encoder_n10` (100% greedy exact match on 50K test samples), SAEs from `sae_encoder_n10`, layer 2 (the critical computation layer; ablating it drops accuracy to 62%).
- **Cylindric**: model `encoder_cyl_10101010_m3` (99.98% exact match), SAEs from `sae_cyl_10101010`, layer 2.
- **Evaluation**: 10,000 test samples (permutation) and 1,000 test samples (cylindric), seeded for reproducibility.
- **Sanity checks**: zeroing all features drops accuracy to 0% (confirming features carry real information); zeroing dead features produces $\Delta \approx 0$.

4 Results

4.1 Permutation RSK

Table **??** shows the top 15 features by causal importance at layer 2.

Of 405 alive features:

- 43 are causally important ($\Delta > 0.1\%$)
- 226 are correlationally strong (lift $> 1.5\times$)
- **1** is both (2%)
- Top-20 overlap between causal and correlational rankings: **0** features

The per-step bumping path correlations are almost entirely epiphenomenal. The most causally important feature (#664, $\Delta = 15.3\%$) has lift 1.11 on step k=1—it fires on step 1's bumping path barely more than on any other P-token. Meanwhile, the 226 features with lift $> 1.5\times$ are individually dispensable: the model routes around any single one.

Table 1: Permutation RSK (layer 2, n=10, 10K test samples). SAE baseline: 100.00% greedy exact match. No feature in the top 30 by causal importance has lift > 1.5.

Feature	Δ Exact	Δ Per-pos	Lift	Best Step	Assessment
664	15.34%	3.07%	1.11	k=1	causal only
895	12.58%	3.03%	1.15	k=9	causal only
361	10.67%	2.87%	1.39	k=1	causal only
387	10.49%	2.91%	1.49	k=0	causal only
678	10.25%	2.21%	1.15	k=7	causal only
657	8.16%	1.67%	1.08	k=7	causal only
230	7.32%	1.94%	1.08	k=4	causal only
406	6.47%	1.61%	1.03	k=2	causal only
409	5.37%	1.46%	1.03	k=3	causal only
23	4.81%	1.21%	1.11	k=7	causal only
564	3.94%	1.34%	1.10	k=8	causal only
129	3.54%	0.88%	1.02	k=3	causal only
611	2.68%	0.71%	1.13	k=8	causal only
753	2.43%	0.49%	1.26	k=8	causal only
715	1.80%	0.40%	1.19	k=1	causal only

4.2 Cylindric Plane Partitions

Table **??** shows the top 15 features by causal importance at layer 2.

Table 2: Cylindric plane partitions (layer 2, profile $(1, 0, 1, 0, 1, 0, 1, 0)$, 1K test samples). SAE baseline: 99.90% exact match. Three of the top four causal features have lift $> 3\times$.

Feature	Δ Exact	Δ Per-pos	Lift	Best Face	Assessment
861	38.10%	3.81%	3.16	f=0	**faithful** ⋆
1006	31.90%	3.21%	3.28	f=6	**faithful** ⋆
469	23.50%	4.56%	1.11	f=3	causal only
177	22.90%	4.14%	1.06	f=6	causal only
38	21.90%	4.23%	1.04	f=4	causal only
279	19.80%	3.94%	1.23	f=2	causal only
185	18.00%	3.43%	1.16	f=6	causal only
294	15.40%	1.54%	3.20	f=6	**faithful** ⋆
129	14.70%	1.91%	1.17	f=1	causal only
746	13.90%	2.23%	1.08	f=0	causal only
91	13.30%	1.36%	1.91	f=3	**faithful** ⋆
865	9.30%	1.99%	1.11	f=4	causal only
230	9.00%	1.03%	1.24	f=6	causal only
516	8.40%	0.89%	1.28	f=2	causal only
551	5.20%	0.54%	1.31	f=1	causal only

Of 356 alive features:

- 49 are causally important ($\Delta > 0.1\%$)
- 267 are correlationally strong (lift $> 1.5\times$)
- **14** are both (29%)
- The three most causally impactful features (#861, #1006, #294) all have lift $> 3\times$ on specific growth diagram faces

Table 3: Summary comparison: compositional faithfulness by task.

	Permutation RSK	Cylindric
Causal features ($\Delta > 0.1\%$)	43	49
Correlated features (lift > 1.5)	226	267
Compositionally faithful	1 (2%)	14 (29%)
Top-20 overlap	0	1
Top causal feature: Δ	15.3%	38.1%
Top causal feature: lift	1.11	3.16

4.3 Comparison

5 Discussion

5.1 Task Constraints Determine Compositional Faithfulness

The central finding is that compositional faithfulness—the alignment between correlational structure and causal importance—depends on the *task*, not the model architecture.

The permutation RSK task admits multiple algorithms: Schensted's sequential bumping and Fomin's parallel growth diagrams. The transformer is free to learn any strategy that inverts the bijection, and the SAE decomposition reveals that it has learned a *distributed* representation where no single feature "owns" a specific algorithmic step. The per-step bumping path correlations, while statistically real (many features have lift $> 1.5\times$), are epiphenomenal—the model can compute the correct output without any individual correlated feature.

The cylindric task admits only one algorithm: the Burge local rule applied face-by-face through a growth diagram (**?**). The computation *must* be localised to faces, so SAE features that correlate with specific faces are also causally necessary. The three most impactful features all have lift $> 3\times$ on identified faces, and zeroing any one of them drops accuracy by 15–38%.

In the language of **?**, the cylindric task constrains the *refinement space*: there are fewer valid decompositions of the computation, so the model's actual syntax is closer to the unique interpretable decomposition. The permutation task leaves the refinement space unconstrained, and the model's chosen syntax does not align with either of the two known human-interpretable algorithms at the single-feature level.

5.2 Implications for Explanatory Optimism

The Weak Principle of Explanatory Optimism (Conjecture 4.3 of **?**) holds that models with implicit explanatory knowledge will tend to represent it in human-understandable form. Our results suggest a refinement: explanatory optimism is *task-conditional*.

When the task structure forces a unique algorithmic solution, the model's internal representation aligns with human-understandable concepts (growth diagram faces), and SAE features are compositionally faithful. When multiple algorithmic strategies are available, the model finds a distributed representation that is computationally effective but not aligned with any single human-understandable algorithm at the piecewise (L2) decomposition level.

This does not refute explanatory optimism outright. The permutation model may still admit a compositionally faithful interpretation at a different refinement level—for instance, Level 3 (global, cross-layer circuits) might reveal that *groups* of features jointly implement bumping steps, even though no single feature does so alone. Our experiment tests L2 faithfulness only.

5.3 Methodological Implications

Our results carry a warning for the mechanistic interpretability community: **correlational SAE feature analysis can be systematically misleading**. In the permutation model, 226 features pass a reasonable correlational threshold (lift $> 1.5\times$), yet essentially none of them are causally important. A research

programme that reports only correlational evidence would conclude that the model has learned Schensted-like bumping steps; the causal evidence shows this conclusion is unfounded.

The fix is straightforward: single-feature ablation is computationally cheap (one forward pass per feature) and should be standard practice alongside correlational analysis. The two measurements are complementary—correlation identifies *what* a feature tracks, causation identifies *whether* it matters—and only their intersection constitutes compositional faithfulness.

5.4 Limitations

- Our analysis is restricted to Level 2 (piecewise, per-layer) refinement. Cross-layer circuits, which may carry the permutation model's true algorithmic structure, require Level 3 analysis.
- The cylindric experiment uses 1,000 test samples (vs 10,000 for permutation), introducing more noise in the Δ estimates.
- We test only layer 2. Other layers may show different patterns of faithfulness.
- The lift threshold of $1.5\times$ is somewhat arbitrary. However, the qualitative result is robust: even at a threshold of $1.2\times$, only 5 of 43 causal features qualify for the permutation model.
- We ablate single features only. Multi-feature ablation (testing whether groups of correlated features are jointly causal) is a natural next step.

6 Related Work

Compositional interpretability. **?** introduce the framework we test. **?** develop the categorical foundations for compositional XAI. **?** propose defining interpretability through symmetry invariance.

Causal methods in interpretability. **?** frame mechanistic interpretability as causal abstraction, using top-down hypothesis testing to find alignments between models and high-level explanations. Our approach is bottom-up: we start from SAE features and test their causal necessity. **?** argue that causal evaluation is essential for interpretability claims to generalise.

SAE interpretability. **?** introduce dictionary learning for transformer decomposition. **?** scale monosemantic feature extraction to Claude 3 Sonnet. **?** connect SAE explanations to minimum description length.

ML for combinatorics. **?** establish the ML4AlgComb benchmark for machine learning on algebraic combinatorics. **?** train the RSK transformer we study here.

7 Conclusion

We provide the first empirical test of the compositional interpretability framework of **?**, using causal feature ablation on SAE decompositions of a combinatorial bijection transformer. The central finding is a task-dependent contrast:

- When the task admits a unique algorithm (cylindric plane partitions), SAE features that correlate with algorithmic steps are also causally necessary. Compositional faithfulness holds: 29% of causal features are also the strongest face-correlated features, and the top three causal features all show $> 3\times$ lift.
- When multiple algorithms exist (permutation RSK), correlational evidence is systematically misleading. The 226 features with strong per-step bumping path correlations are individually dispensable; the 43 causally necessary features show no step-specific structure.

This suggests that explanatory optimism—the conjecture that syntactic compression surfaces human-aligned explanations—is task-conditional. It holds when the task constrains the refinement space to a single decomposition, but fails when the model is free to distribute computation across features in ways that don't map onto known algorithms.

The practical implication is clear: correlational SAE analysis should always be accompanied by causal intervention. Only the intersection of the two constitutes a compositionally faithful interpretation.

Code and models. Source code: `https://github.com/RaggedR/rsk-transformer`. Trained models: `https://huggingface.co/RobBobin/rsk-transformer`.

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